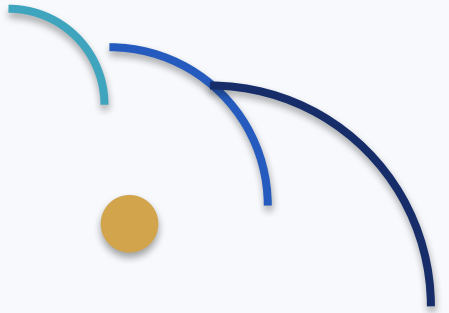
AIORI HACKATHON 2026 | PROJECT ID: PS-014

NETRIVANTA

Pilot Internet-Quality Comparison by Verified Access Medium



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Domain: Smart Cities / Internet Measurement Network

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Problem Description

Why conventional speed tests do not fully describe real-world Internet quality

THE CORE PROBLEM

Traditional Internet speed tests mainly emphasize download and upload speeds, but do not provide a complete picture of Internet quality across access technologies.

PROJECT OBJECTIVE

Build a platform that collects, analyzes and compares multiple Internet-quality parameters across access mediums using real-world measurements.

Limited comparison

Fiber, Cellular, Fixed Wireless and Satellite are not easily compared side by side.

Missing indicators

Latency, packet loss and DNS performance receive limited visibility.

Limited context

Users cannot easily explain differences across locations and networks.

No organized history

Device-specific historical measurement data is difficult to track.

Key idea: measurable evidence of Internet quality — not just advertised speed claims.

Background

Everyday digital experiences depend on more than raw download speed

Online Education

Video lectures freeze; delays disrupt online examinations and interactive sessions.

Remote Work

High latency creates communication delays and interrupts meetings or file transfers.

Online Gaming

Lag and unstable connections affect gameplay even when download speed looks high.

Smart Cities & IoT

Connected devices need reliable, low-latency communication for real-time monitoring.

ISP Selection

Advertised speeds may not reflect everyday performance, making objective comparison difficult.

The common need: understand reliability, responsiveness and performance in real usage conditions.

Literature & RFC Review

Source-based review of the problem context and measurement scope

WHAT THE PROJECT MATERIAL ESTABLISHES

Conventional tests focus mainly on download and upload throughput.

A broader view requires latency, packet-loss and DNS-resolution indicators.

Performance varies with access technology, location, device and network conditions.

Historical records help users examine variation over time.

Measurement scope

Latency • HTTP probe loss • Download speed • Upload speed • Cloudflare DNS-over-HTTPS (DoH) resolution delay

Access-medium comparison

Fiber • Cellular • Fixed Wireless • Satellite

Evidence note: the supplied NETRIVANTA presentation does not list specific papers or RFC identifiers; none have been added here.

Proposed Solution

NETRIVANTA: a web-based Internet-quality measurement and comparison platform

01

User Registration

Create an account and securely sign in.

02

Device Registration

Register multiple devices under one account.

03

Network Measurement

Run Internet performance tests.

04

Parameter Collection

Capture latency, HTTP probe loss, download, upload and DNS DoH delay.

05

Data Storage

Associate each measurement with the correct user and device.

06

Historical Analysis

Review previous measurements and compare performance over time.

Outcome: one structured workflow connecting accounts, devices, measurements and historical comparisons.

Solution Optimization

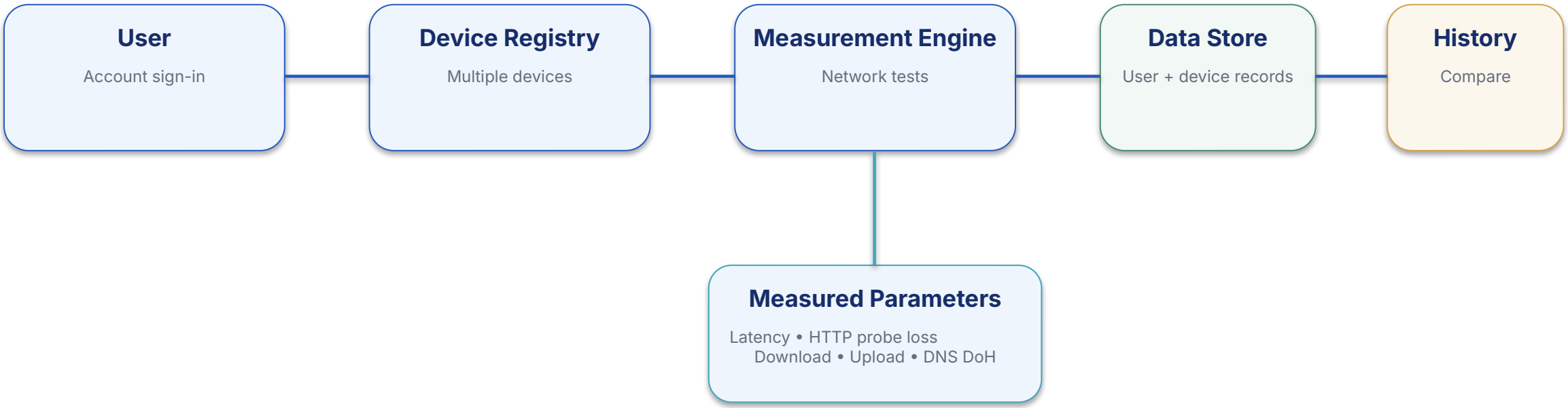
How NETRIVANTA broadens the view beyond conventional speed testing

Aspect	Conventional speed-test view	NETRIVANTA approach
Primary focus	Download and upload speed	Multi-parameter Internet quality
Network indicators	Throughput emphasis	Latency, HTTP probe loss, download, upload, DNS DoH delay
Device context	Not emphasized in source problem framing	Multiple registered devices under one account
History	Limited visibility in source problem framing	Device-wise and account-wise measurement history
Comparison	Difficult across access technologies	Comparison-oriented collection across four access mediums

Optimization principle: combine multiple indicators with device context and historical records for more informed comparison.

Solution Architecture & Design (UML View)

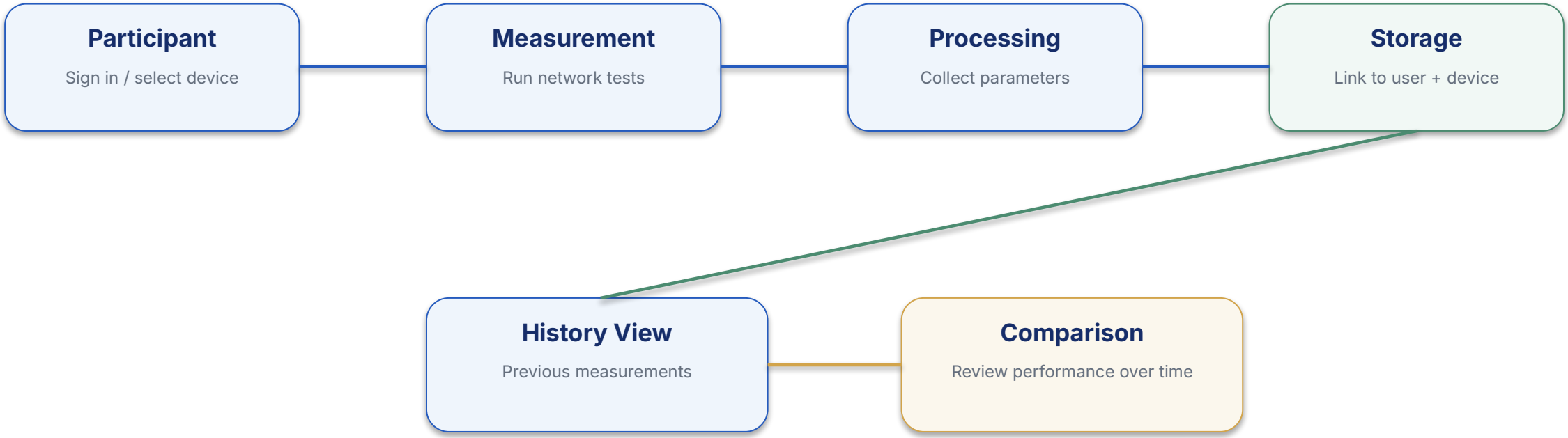
Conceptual system flow derived from the workflow described in the NETRIVANTA presentation



Architecture emphasizes secure account handling, device identification, reliable measurements and structured performance analysis.

Data Flow Diagram (DFD)

Information flow from participant action to historical comparison



Data captured: latency • HTTP probe loss • download speed • upload speed • Cloudflare DNS-over-HTTPS resolution delay

Deployment Timeline

Suggested implementation sequence based on the platform workflow

1

PHASE 1

Account setup

User registration and secure sign-in

2

PHASE 2

Device onboarding

Register and associate multiple devices

3

PHASE 3

Measurement

Run tests and collect all defined parameters

4

PHASE 4

Data & history

Store user/device-linked records and enable history

5

PHASE 5

Evaluation

Review measurements and compare access mediums

Note: the source presentation describes platform capabilities but does not provide dated milestones; this is a proposed sequence, not a reported schedule.

Bibliography & Source Material

References used to prepare this presentation

Primary project source

NETRIVANTA — Pilot Internet-Quality Comparison by Verified Access Medium. AIORI Hackathon 2026, Project ID PS-014. Presentation supplied by the project team.

Project demonstration URL

https://netrivanta.onrender.com

Research limitation: the supplied project presentation does not identify external research papers, RFCs or standards. No external citations have been fabricated or added.

All diagrams in this deck are conceptual visualizations of the workflow described in the supplied source.

Thank You

NETRIVANTA

Measure Beyond Speed. Understand Internet Quality.

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